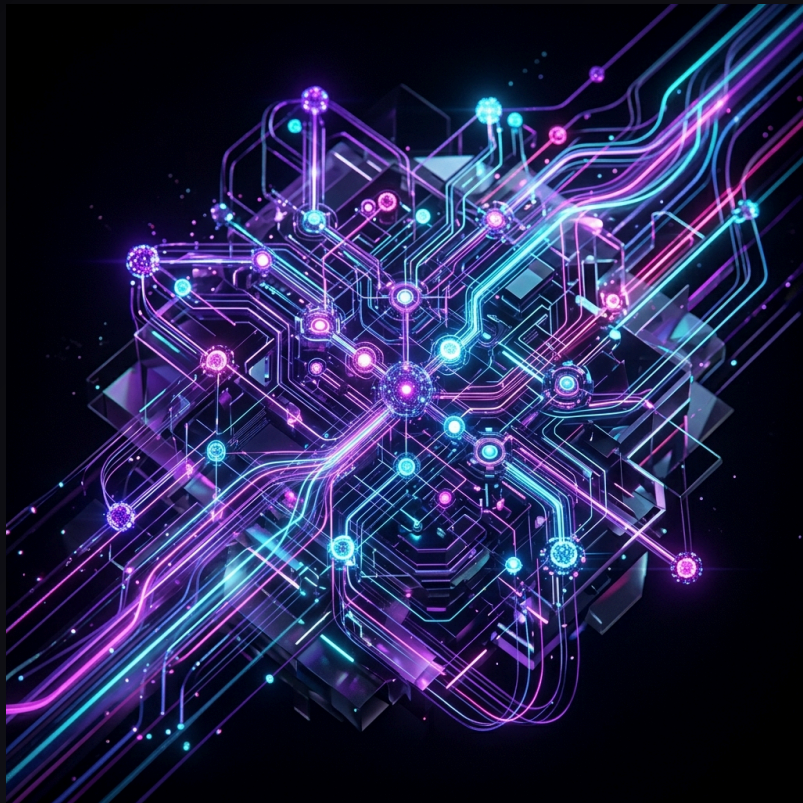


SkillSage

AI-Powered Technical Assessment Platform



A resume tells you what a candidate claims to know. SkillSage conducts adaptive, real-time technical interviews to prove exactly how well they know it.

Deployed URL: <https://thriving-elegance-production.up.railway.app/>

GitHub: <https://github.com/mijwad7/skill-sage.git>

Architecture & System Flow



1. Data Intake & Structuring

Skill Extractor: Parses unstructured text from the Job Description and Candidate Resume to map out a structured array of required skills versus claimed skills (with confidence scores).

2. Dynamic Evaluation

Gap Analyzer & Assessment Agent: Compares requirements against claims to build an interview queue. The Agent conducts a dynamic interview, adapting the difficulty of each question (easy → medium → hard) based on real-time semantic evaluation of the user's answers.

3. Final Synthesis

Scoring, Verification, & Plan Generation: The system aggregates scores (0-100), flags mismatched claims vs. reality (Verification Engine), and uses an AI manager prompt to generate a final Hire/No-Hire recommendation alongside a personalized learning roadmap.

Scoring & Agent Logic

How Adaptive Questioning Works

The AI does not rely on a static question bank. Instead, every answer is fed back into the LLM with the context of the target skill. If the candidate gives a basic answer, the Agent generates a follow-up question that increases in complexity (e.g. moving from definitions to architecture or edge cases). This mimics a real senior engineering interview.

Score Band Explanation

The semantic evaluation engine assigns a deterministic 1-10 score per answer:

- **1-3 (Beginner):** Incorrect or superficial.
- **4-6 (Developing):** Partially correct; missing key concepts.
- **7-8 (Proficient):** Mostly correct; solid practical understanding.
- **9-10 (Expert):** Comprehensive, well-reasoned, understands trade-offs.

These are aggregated across all questions for a skill to produce a definitive 0-100% mastery score.

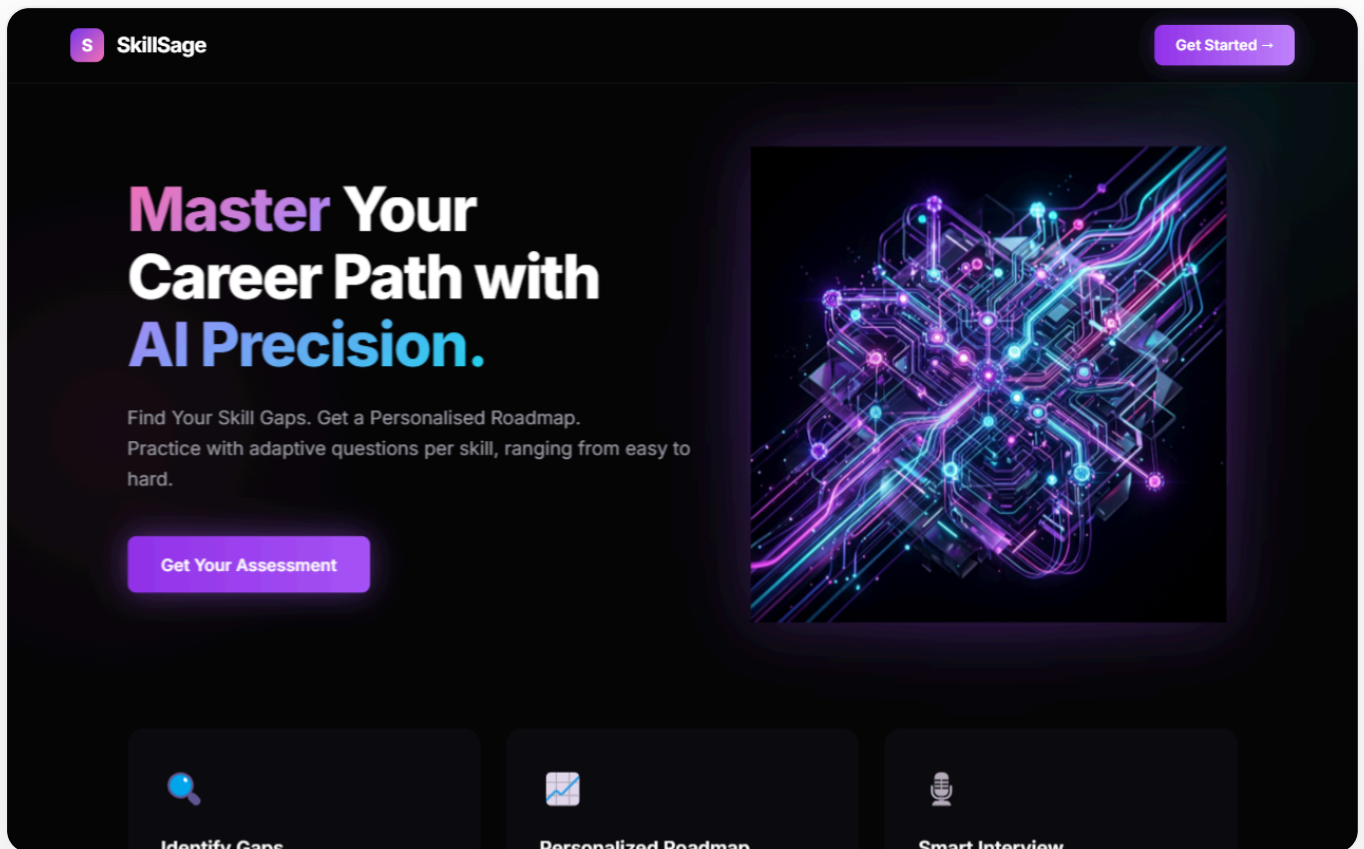
Claim vs. Reality Logic (Verification Engine)

SkillSage uniquely exposes candidates who over-index on their resumes. The logic relies on crossing extracted resume confidence with final assessment scores:

- *If Resume Claim > 75% AND Score < 45% → ⚠ Inflated Claim*
- *If Resume Claim < 30% AND Score > 80% → ★ Undervalued Strength*
- *If JD Requirement AND Resume Claim < 30% AND Score < 50% → ❌ Hidden Gap*

Application Screenshots

1. Setup & Intelligent Processing



2. Adaptive AI Interview Interface

PROGRESS & SKILLS

CSS-in-JS · 06/6

Tailwind CSS

React Testing Library

TypeScript

Next.js

GraphQL

DIFFICULTY METER



ADAPTIVE LEVEL: **Easy**

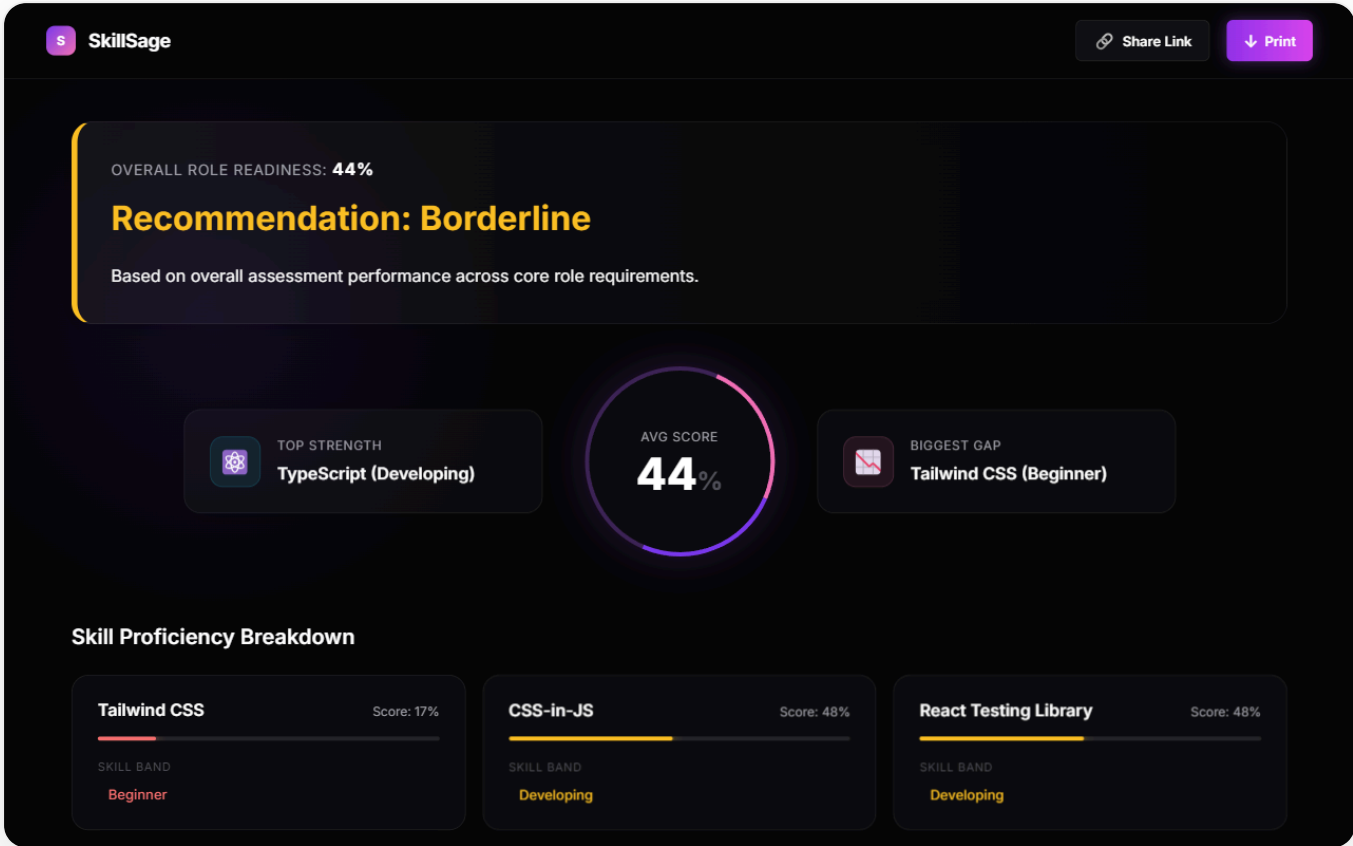


Type your response here...

Send ↗

Application Screenshots (Cont.)

3. Results, Verification & Recommendations



Sample Agent Input / Output

SkillSage operates by enforcing strict structured JSON outputs from the Google Gemini Flash model. Below is a realistic example of the Agent's internal payload after evaluating a candidate's answer.

Candidate Input

```
"I use React to build components. I use useState to keep track of state variables."
```

AI Agent Output Payload

```
{
  "evaluation": {
    "score": 4,
    "feedback": "You mentioned useState, but missed how you handle side effects or global state coordination.",
    "key_gap": "useEffect and Context API/Redux",
    "insight": "Candidate demonstrated basic hook knowledge. Escalating to side-effect management to test intermediate proficiency."
  },
  "next_question": "If you needed to fetch data from an external API when the component first loads, how would you implement that in React?",
  "difficulty_adjustment": "medium"
}
```

Notice the "Insight" trace: This string is passed directly to the frontend to visually show the user exactly why the AI decided to ask the next question, building trust in the platform's adaptive capabilities.

Innovation Highlights



Voice Interview Integration

Utilizing native Web Speech APIs, the platform supports seamless Text-to-Speech (TTS) and Speech-to-Text (STT). Candidates can dictate their answers hands-free, simulating a real human interview environment.



Verification Insights

Rather than just generating a score, the platform mathematically proves the delta between what a candidate claims on paper versus what they can actually execute, solving the core problem of resume embellishment.



Recruiter Share Links

Secure, unguessable UUID-based routing allows candidates to instantly share their verified scorecards and roadmaps with recruiters without requiring account creation or complex auth gateways.



AI Hire Recommendation

The platform cross-references the entire session's performance against the original Job Description to output a definitive "Role Readiness" score and an HR-ready hiring recommendation (e.g., "Hire with Targeted Upskilling").